

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Designed and executed **experimental test strategies** for reagent systems and assays, focusing on **product development** for new diagnostic instruments.
- Analysed complex **experimental data** using statistical methods and presented key findings to cross-functional review boards, informing critical **product decisions**.
- Investigated and resolved technical issues in laboratory workflows, applying **sound scientific judgment** to ensure robust **assay development** and product quality.
- Developed and refined **system-level requirements** for new assays, aligning technical specifications with overarching program and product goals for **clinical laboratory science**.
- Prepared comprehensive **technical reports** and study summaries detailing laboratory findings in compliance with **good laboratory practice (GLP)** standards.
- Collaborated with **cross-functional teams** including R&D, manufacturing, and quality assurance to meet program achievements and ensure timely product delivery.

SKILLS

Laboratory Science – Experimental Design, Assay Development, GLP, Reagent Systems, Workflow Optimization, Troubleshooting

Data Analysis & Statistics – Statistical Analysis, Data Interpretation, Sample Size Calculation, Statistical Power Estimation, R, Python (pandas, NumPy, scikit-learn)

Scientific Communication – Technical Reports, Study Summaries, Presentation Skills, Oral Communication, Written Communication

Product Development – System-Level Requirements, Product Goals Alignment, Regulatory Requirements Integration, Quality Assurance

Tools & Technologies – Power BI, Tableau, SQL, Jira, Confluence, Linux

PROJECTS

Development of a Novel Diagnostic Assay Workflow

- Designed and implemented a multi-stage experimental plan to validate a new diagnostic assay, integrating **reagent system** performance metrics.
- Performed rigorous **statistical analysis** on experimental output, identifying key performance indicators and recommending workflow optimizations that improved assay consistency by **18%**.
- Authored detailed **technical documentation** and presented study outcomes to a scientific review committee, influencing the go/no-go decision for pilot production.

Automated Data Analysis Pipeline for Scientific Studies

- Engineered a **Python-based data processing pipeline** to automate the analysis of large experimental datasets, reducing manual processing time by approximately **60%**.
- Developed interactive **Power BI dashboards** to visualize assay performance trends and troubleshooting insights for research and development teams.
- Integrated custom scripts with existing **SQL databases** to extract and transform raw experimental data, ensuring data integrity for downstream analysis.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)